BAYESIAN MODELING OF COMPLEXITY

Sensitivity Analysis & Robustness

Meysam Goodarzi
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Hertie School of Public Governance

Agenda

1 Racap & Today

2 Sensitivity Analysis
- Global Sensitivity
- Local Sensitivity

3 Robustness Analysis

4 Wrap-up & Next

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Recap

Session 7: Bayesian Linear Models

- Bayesian regression: Prior $\times$ Likelihood $\rightarrow$ Posterior
- Priors as regularization (Ridge)
- Full posterior distribution over β and σ (not just point estimates)
- Model quality
- Implementation in PyMC

We can now fit Bayesian regression models!

Today's question: But should we trust our models?

Model Assessment

Fitting a model is easy, validating it is hard.

Why model assessment matters:

- Models are simplifications $\rightarrow$ always wrong to some degree
- Need to know *how* and *where* they fail
- Bad models lead to bad decisions (policy, predictions)
- Models can be in "garbage in, garbage out" state

"All models are wrong, but some are useful."

Today's goal: Learn to assess the sensitivity and robustness

What is Sensitivity Analysis?

Question: How do model conclusions depend on assumptions?

Three types:

1. **Global**: Effect of inputs across the entire parameter space ($\theta = [\mu, \sigma, ...]$)
2. **Local**: Effect of small parameter changes near baseline ($\mu \pm \epsilon$)
3. **Robustness**: Effect of modeling assumptions (type of prior, e.g., Normal, Cauchy, etc.)

Why it matters:

- **Scientific understanding**: Which factors matter the most?
- **Model validation**: Are the results robust to assumptions?
- **Resource allocation**: Where to collect more data?

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Global Sensitivity Analysis

Goal: Understand the effect of inputs across the full parameter space

Approach:

1. Vary inputs over plausible ranges
2. Run model for each input combination
3. Analyze input-output relationship

Methods:

- **Test stats**: Effect of parameter changes on the test stats
- **Posterior**: Marginal effect of each input on the posterior

For Bayesian models:

- Vary priors over reasonable range
- Check if posterior conclusions change
- If stable $\rightarrow$ conclusions robust
- If sensitive $\rightarrow$ need stronger prior or more data

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Local Sensitivity Analysis

Goal: Understand the sensitivity of the posterior (or a test statistic) around a parameter

Approach:

- Compute gradients $\alpha_j = s_j = \frac{\partial y}{\partial x_j}$

  Interpretation: Unit change in $x_j \rightarrow s_j$ change in y
- Measure elasticities $\omega_j = \frac{\partial \log y}{\partial \log x_j}$

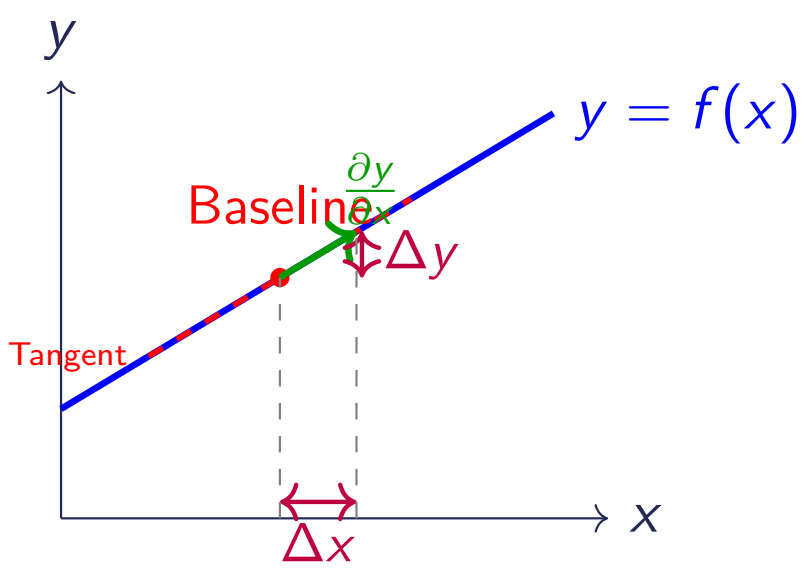
  Interpretation: 1% change in $x_j \rightarrow \omega_j$% change in y

Use: Understanding how much are the results reliable in case there is a change in the parameters of the model.

Local Sensitivity: Visual

Local sensitivity:
Slope of tangent line at baseline point

$$\frac{\Delta y}{\Delta x} \approx \frac{\partial f}{\partial x}$$



When local is sufficient:

- Policy changes near status quo
- Small perturbations expected
- Linear region

When is global needed:?

- Large parameter uncertainty
- Interactive terms in the model
- Full parameter space exploration

Prior Sensitivity

Test: How do conclusions change with prior parameters?

Procedure: Compare posterior means and intervals after

1. Fitting model with baseline prior (e.g., $\beta \sim \mathcal{N}(0, 5^2)$)
2. Refitting with wider prior (e.g., $\beta \sim \mathcal{N}(0, 10^2)$)
3. Refitting with narrower prior (e.g., $\beta \sim \mathcal{N}(0, 2^2)$)

Interpretation:

- **Results similar**: Data dominate, conclusions robust
- **Results change substantially**: Prior matters, need:
  - More data, or
  - Better prior (from domain knowledge), or
  - Acknowledge sensitivity in conclusions

Rule of thumb:
Small samples ($n < 50$): Prior matters
Large samples ($n > 200$): Data dominates
Always check with robustness analysis!

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Robustness to Model Assumptions

Question: Are conclusions robust to modeling choices?

Common robustness checks:

1. **Likelihood family:**
   - Normal vs. Student-t (heavy tails)
   - Check if outliers dominate conclusions
2. **Prior specification:**
   - Widen/narrow prior scales
   - Try different prior families
   - If results stable $\rightarrow$ robust
3. **Model structure:**
   - Linear vs. non-linear
   - Include/exclude interactions
   - Different transformations
4. **Data subsets:**
   - Remove influential observations
   - Temporal stability (early vs. late data)
   - Spatial stability (different regions)

Student-t for Robustness

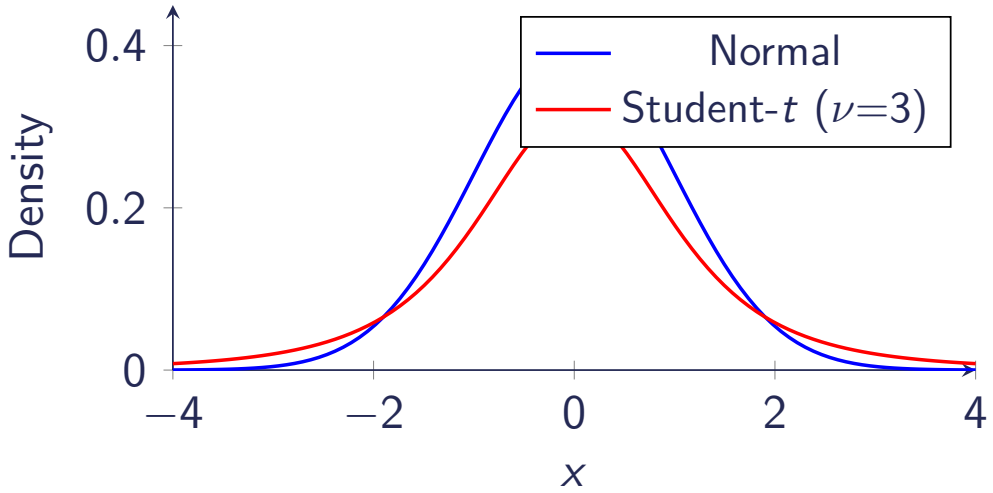
Problem: Normal likelihood sensitive to outliers

Solution: Use Student-t distribution

Student-t regression: $y_i \mid \mathbf{x}_i, \beta, \sigma, \nu \sim t_\nu(\mathbf{x}_i^T \beta, \sigma^2)$

Parameters:

- ν = degrees of freedom
- Small ν (e.g., 3-5) $\rightarrow$ heavy tails $\rightarrow$ robust to outliers
- Large ν (e.g., 30+) $\rightarrow$ approaches Normal



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Key Concepts

Posterior predictive checks:

- Generate data from model
- Compare to observed data
- Visual and quantitative tests

Diagnostics:

- Residual plots reveal misspecification
- Check for patterns, outliers, heteroscedasticity
- Influential observations can drive conclusions

Model comparison:

- WAIC and LOO balance fit and complexity
- Use $2\times$SE rule for significant differences

Sensitivity analysis:

- Global sensitivity
- Local sensitivity
- Robustness

Questions?

Next: Bayesian Hierarchical Models!